INDIVIDUAL ACTIVITY — 60 MINUTES

MIS Integration Impact Analysis

Information Technology – MIS | Understanding Major Functional Systems & ROI of MIS Implementation

The Scenario

TechNova Industries is a mid-sized smart-home devices manufacturer in Pune with 350 employees. They currently operate with **standalone, disconnected systems** across departments — spreadsheets for inventory, manual ledgers for finance, paper-based HR records, and no integrated sales tracking.

The management is considering implementing an **integrated MIS** (Enterprise Resource Planning system) that connects all five major functional areas: **Marketing & Sales, Finance & Accounting, Manufacturing & Production, Human Resource, and Inventory**. As an independent business analyst, **you** must calculate the before-and-after metrics for each functional system, quantify the total savings, and determine whether the MIS investment is financially justified.

Assumption: Sufficient market demand exists to sell all additional units produced after MIS implementation.

Reference Data Table

Use **red values** for current (before MIS) state, **green values** for proposed (after MIS) state. All values are given — do not estimate.

PARAMETER	VALUE
MARKETING & SALES SYSTEM	
Monthly Leads Generated	600 → 900
Lead-to-Customer Conversion Rate	3.0% → 5.5%
Average Order Value	₹12,000 → ₹14,500
Number of Sales Representatives	15 (unchanged)

PARAMETER	VALUE
Monthly Marketing Spend	₹6,00,000 → ₹7,50,000
FINANCE & ACCOUNTING SYSTEM	
Monthly Invoices Processed	800 (unchanged)
Processing Cost per Invoice	₹220 → ₹75
Invoice Error Rate	5.0% → 0.6%
Error Correction Cost (per error)	₹3,000
Annual Revenue	₹10,00,00,000
Days Sales Outstanding (DSO)	48 days → 30 days
Cost of Capital	12% per annum
MANUFACTURING & PRODUCTION SYSTEM	
Daily Production Capacity	300 units
Actual Daily Output	225 units → 275 units
Defect Rate	4.0% → 1.5%
Manufacturing Cost per Unit	₹850 → ₹780
Selling Price per Unit	₹1,500
Working Days per Year	300 days
HUMAN RESOURCE SYSTEM	
Total Employees	350 (unchanged)
Annual Employee Turnover Rate	24% → 15%
Cost per Employee Replacement	₹95,000 → ₹70,000
Annual Training Cost per Employee	₹18,000 → ₹11,000
INVENTORY SYSTEM	
Annual Cost of Goods Sold (COGS)	₹6,00,00,000
Average Inventory Value	₹2,00,00,000 → ₹1,00,00,000
Inventory Carrying Cost Rate	28% → 20%

PARAMETER	VALUE
Monthly Stockout Incidents	12 → 3
Cost per Stockout Incident	₹25,000
MIS INVESTMENT	
Software Licenses (One-time)	₹60,00,000
Hardware & Infrastructure (One-time)	₹25,00,000
Implementation & Training (One-time)	₹20,00,000
Annual Maintenance & Support	₹12,00,000

√× Key Formulas to Remember

`CAC = Marketing Spend ÷ New Customers` | `OEE = (Actual Output ÷ Capacity) × (1 − Defect Rate)`

`Inventory Turnover = COGS ÷ Avg Inventory` | `Capital Freed = DSO Reduction × (Annual Revenue ÷ 365)`

`Payback (months) = (Total Investment ÷ Net Annual Savings) × 12`

Part A: Marketing & Sales System

Calculate revenue metrics and customer acquisition efficiency before and after MIS. The MIS provides CRM-driven lead scoring, automated follow-ups, and cross-sell recommendations.

A1 Calculate the **Monthly Revenue** before and after MIS implementation.

Before: ₹XX,XX,XXX

Before MIS (₹/month)

After: ₹XX,XX,XXX

After MIS (₹/month)

Show Hint

Formula: Monthly Revenue = Monthly Leads × Conversion Rate × Average Order Value

Before: 600 × 3.0% × ₹12,000 = 600 × 0.03 × 12,000
After: 900 × 5.5% × ₹14,500 = 900 × 0.055 × 14,500

A2 Calculate the **Revenue per Sales Rep per Month** before and after MIS.

Before: ₹XX,XXX

Before MIS (₹/rep/month)

After: ₹XX,XXX

After MIS (₹/rep/month)

Show Hint

Simply divide your A1 answers by 15 (number of sales reps). This shows how MIS makes each salesperson more productive without adding headcount.

A3 Calculate the **Customer Acquisition Cost (CAC)** before and after MIS. Is the higher marketing spend after MIS justified?

Before: ₹XX,XXX

Before MIS (₹/customer)

After: ₹XX,XXX

After MIS (₹/customer)

Show Hint

CAC = Marketing Spend ÷ Number of New Customers

New Customers = Leads × Conversion Rate
Before: ₹6,00,000 ÷ (600 × 0.03) = ₹6,00,000 ÷ 18
After: ₹7,50,000 ÷ (900 × 0.055) = ₹7,50,000 ÷ 49.5

Even though marketing spend increased by ₹1.5L, the CAC should drop

significantly because MIS generates far more conversions per rupee spent.

Part B: Finance & Accounting System

B

Calculate processing efficiency, error reduction, and working capital impact. The MIS automates invoice generation, reconciliation, and payment tracking.

B1 Calculate the **Monthly Invoice Processing Cost** before and after MIS.

Before: ₹X,XX,XXX

Before MIS (₹/month)

After: ₹XX,XXX

After MIS (₹/month)

Show Hint

Processing Cost = Number of Invoices × Cost per Invoice

Before: 800 × ₹220
After: 800 × ₹75

MIS automates data entry, validation, and routing, dramatically reducing per-invoice cost.

B2 Calculate the **Monthly Error Correction Cost** before and after MIS.

Before: ₹X,XX,XXX

Before MIS (₹/month)

After: ₹XX,XXX

After MIS (₹/month)

Show Hint

Error Cost = Invoices × Error Rate × Cost per Error Correction

Before: 800 × 5.0% × ₹3,000 = 800 × 0.05 × 3,000
After: 800 × 0.6% × ₹3,000 = 800 × 0.006 × 3,000

Automated validation in MIS catches errors before invoices are sent.

B3 **Faster payment collection reduces DSO. Calculate the Annual Interest Savings from working capital freed up by the DSO improvement.**

e.g. ₹XX,XXX

Unit: Rupees per year (interest saved on freed capital)

Show Hint

Step 1: Daily Revenue = Annual Revenue ÷ 365 = ₹10,00,00,000 ÷ 365
Step 2: DSO Reduction = 48 − 30 = 18 days
Step 3: Working Capital Freed = Daily Revenue × DSO Reduction
Step 4: Annual Interest Savings = Capital Freed × Cost of Capital (12%)

The MIS automates follow-ups and payment reminders, so customers pay faster — freeing cash that would otherwise be tied up in receivables.

C

Part C: Manufacturing & Production System

Calculate production efficiency, quality improvement, and additional profit from increased output. The MIS provides real-time shop-floor scheduling, predictive maintenance alerts, and quality control tracking.

C1 **Calculate the Overall Equipment Effectiveness (OEE) before and after MIS.**

Before: XX.X%

Before MIS (%)

After: XX.X%

After MIS (%)

Show Hint

OEE = (Actual Output ÷ Capacity) × (1 − Defect Rate)

This combines utilization (output vs. capacity) with quality (good units vs. total produced).

Before: (225 ÷ 300) × (1 − 0.04) = 0.75 × 0.96
After: (275 ÷ 300) × (1 − 0.015) = 0.9167 × 0.985

An OEE below 65% is considered "poor" in manufacturing. World-class is 85%+.

C2 Calculate the **Annual Cost of Defective Units** before and after MIS (defective units are scrapped at full unit cost).

Before: ₹XX,XX,XXX

Before MIS (₹/year)

After: ₹XX,XX,XXX

After MIS (₹/year)

Show Hint

Defect Cost = Daily Output × Defect Rate × 300 days × Unit Cost

Before: 225 × 4.0% × 300 × ₹850 = 225 × 0.04 × 300 × 850
After: 275 × 1.5% × 300 × ₹780 = 275 × 0.015 × 300 × 780

Note: Use the respective unit cost for each period (₹850 before, ₹780 after) since MIS also reduces per-unit cost through better material planning.

C3 With higher output and lower defects, more good units are produced. Calculate the **Additional Annual Profit** from the extra good units sold (Selling Price − Unit Cost = profit per unit; use post-MIS unit cost).

e.g. ₹X,XX,XX,XXX

Unit: Rupees per year (additional profit)

Show Hint

Step 1: Good units before = 225 × (1 − 0.04) × 300 days
Step 2: Good units after = 275 × (1 − 0.015) × 300 days
Step 3: Additional good units = After − Before
Step 4: Profit per additional unit = ₹1,500 − ₹780 (post-MIS cost)
Step 5: Additional Profit = Additional good units × Profit per unit

This captures the MIS's biggest financial impact: better scheduling and less downtime = more products sold.

Part D: Human Resource System

D

Calculate turnover costs and training expenses. The MIS provides employee self-service portals, automated payroll, performance tracking, and streamlined recruitment workflows.

D1 Calculate the **Annual Employee Turnover Cost** before and after MIS.

Before: ₹XX,XX,XXX

Before MIS (₹/year)

After: ₹XX,XX,XXX

After MIS (₹/year)

Show Hint

Turnover Cost = Employees × Turnover Rate × Replacement Cost

Before: 350 × 24% × ₹95,000
After: 350 × 15% × ₹70,000

TIME LEFT 60:00 Start Reset

Note: 350 × 15% = 52.5 employees (annual average, can be fractional).

D2 Calculate the **Total Annual Training Cost** for all 350 employees before and after MIS.

Before: ₹XX,XX,XXX

Before MIS (₹/year)

After: ₹XX,XX,XXX

After MIS (₹/year)

Show Hint

Total Training Cost = Number of Employees × Training Cost per Employee

Before: 350 × ₹18,000
After: 350 × ₹11,000

MIS enables e-learning platforms, automated training schedules, and digital certification — reducing per-employee training cost by 39%.

D3 What is the **Total Annual HR Savings** after MIS implementation (turnover cost reduction + training cost reduction combined)?

e.g. ₹XX,XX,XXX

Unit: Rupees saved per year

Show Hint

Total HR Savings = (Turnover Cost Before − Turnover Cost After) + (Training Cost Before − Training Cost After)

Add the two savings figures together. This is the annual HR department budget relief from MIS.

Part E: Inventory System

E

Calculate inventory efficiency metrics. The MIS provides real-time stock visibility, automated reorder points, demand forecasting, and warehouse optimization.

E1 **Calculate the Inventory Turnover Ratio before and after MIS.**

Before: X.XX

Before MIS (ratio)

After: X.XX

After MIS (ratio)

Show Hint

Inventory Turnover = COGS ÷ Average Inventory Value

Before: ₹6,00,00,000 ÷ ₹2,00,00,000
After: ₹6,00,00,000 ÷ ₹1,00,00,000

A higher ratio means inventory moves faster. MIS cuts average inventory in half by eliminating overstocking and improving demand prediction.

E2 **Calculate the Annual Inventory Carrying Cost before and after MIS.**

Before: ₹XX,XX,XXX

Before MIS (₹/year)

After: ₹XX,XX,XXX

After MIS (₹/year)

Show Hint

Carrying Cost = Average Inventory Value × Carrying Cost Rate

Before: ₹2,00,00,000 × 28%
After: ₹1,00,00,000 × 20%

Carrying cost includes warehousing, insurance, obsolescence risk, and opportunity cost of capital tied up in inventory. MIS reduces both the inventory value AND the carrying rate (better warehouse utilization).

E3 Calculate the **Annual Stockout Cost** before and after MIS.

Before: ₹XX,XX,XXX

Before MIS (₹/year)

After: ₹XX,XX,XXX

After MIS (₹/year)

Show Hint

Annual Stockout Cost = Monthly Incidents × Cost per Incident × 12 months

Before: 12 × ₹25,000 × 12

After: 3 × ₹25,000 × 12

Stockout costs include lost sales, expedited shipping, and customer dissatisfaction. MIS's demand forecasting and automated reordering reduce stockouts by 75%.

Part F: The MIS Business Case

This is where all five functional systems come together. Consolidate your calculations to determine whether the MIS investment is financially justified. Refer back to ALL your answers from Parts A through E.

F1 Calculate the **Total Annual Savings/Benefits** from MIS across all five functional systems. Sum up: (a) Marketing revenue increase, (b) Invoice processing savings, (c) Error cost savings, (d) Working capital interest savings, (e) Defect cost reduction, (f) Additional production profit, (g) HR savings, (h) Inventory carrying cost savings, (i) Stockout cost savings.

e.g. ₹X,XX,XX,XXX

Unit: Total Rupees saved/earned per year (before deducting MIS costs)

Show Hint

Go through each component and annualize (multiply by 12) if your answer was monthly:

(a) (A1 After − A1 Before) × 12 — Marketing revenue increase
(b) (B1 Before − B1 After) × 12 — Invoice processing savings
(c) (B2 Before − B2 After) × 12 — Error cost savings
(d) B3 answer — Working capital interest (already annual)
(e) C2 Before − C2 After — Defect cost reduction (already annual)
(f) C3 answer — Additional production profit (already annual)
(g) D3 answer — Total HR savings (already annual)
(h) E2 Before − E2 After — Carrying cost savings (already annual)
(i) E3 Before − E3 After — Stockout cost savings (already annual)

Add all nine components together for the total.

F2 **Calculate the Payback Period in months. First compute the one-time investment (software + hardware + implementation). Then subtract the annual maintenance cost from F1 to get net annual savings. Finally, calculate how many months until the investment is recovered.**

e.g. XX.X months

Unit: Months (round to 1 decimal place)

Show Hint

Step 1: One-time Investment = ₹60,00,000 + ₹25,00,000 + ₹20,00,000 = ?
Step 2: Net Annual Savings = F1 − ₹12,00,000 (annual maintenance)
Step 3: Payback (months) = (One-time Investment ÷ Net Annual Savings) × 12

If the result is under 12 months, the MIS pays for itself in less than a year!

F3 **Calculate the 3-Year Net Benefit — the total money gained over 3 years after fully recovering the one-time investment.**

e.g. ₹X,XX,XX,XXX

Unit: Rupees (net gain over 3 years)

Show Hint

Step 1: 3-Year Gross Savings = Net Annual Savings (from F2 Step 2) × 3
Step 2: 3-Year Net Benefit = 3-Year Gross Savings − One-time Investment

This shows the *true* financial return over a typical MIS lifecycle period. Compare this to the initial investment to see the multiplier effect.

Reveal Complete Solution

Click once to arm, click again to reveal. Use only after attempting all questions.